

Concise review: Adult salivary gland stem cells and a potential therapy for xerostomia.

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The ability to speak, swallow, masticate, taste food, and maintain a healthy oral cavity is heavily reliant on the presence of saliva, the hugely important effect of which on our everyday lives is often unappreciated. Hyposalivation, frequently experienced by people receiving radiation therapy for head and neck cancers, results in a plethora of symptoms whose combined effect can drastically reduce quality of life. Although artificial lubricants and drugs stimulating residual function are available to ameliorate the consequences of hyposalivation, their effects are at best transient. Such management techniques do not address the source of the problem: a lack of functional saliva-producing acinar cells, resulting from radiation-induced stem cell sterilization. Post-radiotherapy stimulation of cell proliferation only results in improved saliva secretion when part of the tissue has been spared or when the dose to the salivary gland (SG) remains below a certain level. Therefore, stem cell replacement therapy may be a good option to treat radiation-induced hyposalivation. Substantial progress has been made lately in the understanding of cell turnover in the SG, and the recent identification of stem and progenitor cell populations in the SG provides a basis for studies toward development of a stem cell-based therapy for xerostomia. Here, we review the current state of knowledge of SG stem cells and their potential for use in a cell-based therapy that may provide a more durable cure for hyposalivation.

Neurotrophic factor GDNF promotes survival of salivary stem cells.

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Stem cell-based regenerative therapy is a promising treatment for head and neck cancer patients that suffer from chronic dry mouth (xerostomia) due to salivary gland injury from radiation therapy. Current xerostomia therapies only provide temporary symptom relief, while permanent restoration of salivary function is not currently feasible. Here, we identified and characterized a stem cell population from adult murine submandibular glands. Of the different cells isolated from the submandibular gland, this specific population, Lin-CD24+c-Kit+Sca1+, possessed the highest capacity for proliferation, self renewal, and differentiation during serial passage in vitro. Serial transplantations of this stem cell population into the submandibular gland of irradiated mice successfully restored saliva secretion and increased the number of functional acini. Gene-expression analysis revealed that glial cell line-derived neurotrophic factor (Gdnf) is highly expressed in Lin-CD24+c-Kit+Sca1+ stem cells. Furthermore, GDNF expression was upregulated upon radiation therapy in submandibular glands of both mice and humans. Administration of GDNF improved saliva production and enriched the number of functional acini in submandibular glands of irradiated animals and enhanced salisphere formation in cultured salivary stem cells, but did not accelerate growth of head and neck cancer cells. These data indicate that modulation of the GDNF pathway may have potential therapeutic benefit for management of radiation-induced xerostomia.

Adipose derived or bone-marrow derived mesenchymal stem cell treatment for hyposalivation: protocol for a systematic review and network meta-analysis.

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Salivary hypofunction leads to debilitating oral symptoms and has major complications for overall quality

of life. Two of the most frequent causes of xerostomia are radiotherapy in the head and neck and Sjögren's syndrome. Only symptomatic treatment is available today. An increasing number of both preclinical and clinical studies have suggested that mesenchymal stem cell (MSC) transplantation treatment can increase the salivary flow rate and ameliorate symptoms of xerostomia. However, both adipose-derived and bone marrow-derived MSCs are used, although they differ in important ways. The primary objective of this study is an indirect comparison of the change in the unstimulated salivary flow rate after intervention between patients treated with adipose-derived or bone marrow-derived MSCs. This systematic review and network meta-analysis will search for eligible studies in the MEDLINE, EMBASE, and Cochrane CENTRAL register of Controlled Trials. Eligible studies are as follows: clinical studies including human patients with salivary hypofunction due to either radiotherapy or Sjogren's syndrome who were subsequently treated with either adipose-derived MSCs or bone marrow-derived MSCs. Studies with no control group will be excluded. The search phrase has been peer-reviewed following the PRESS guidelines. The primary outcome is the change in the unstimulated salivary flow rate after treatment with either adipose-derived or bone marrow-derived MSCs. Secondary outcomes are as follows: change in patient reported outcomes, methods of intervention administration, number of injected MSCs, and safety. Data from included studies will be pooled and compared with a fixed-effects or random effects model dependent on signs of heterogeneity, presented with a forest plot, and indirectly compared with a meta-regression in a network meta-analysis. Risk of bias will be assessed with the tools ROBINS-I or RoB-2 depending on type of study. Both adipose-derived and bone marrow-derived MSCs are used today for experimental treatment of salivary hypofunction in humans as no direct or indirect comparisons have been made. Therefore, an evaluation of the effect of adipose-derived vs bone marrow-derived MSC treatment is needed to support future decision-making on the type of MSC used in a clinical trial. PROSPERO ID CRD42024527183.

Salivary gland organoid transplantation as a therapeutic option for radiation-induced xerostomia.

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Xerostomia is a pathological condition characterized by decreased salivation due to salivary gland dysfunction and is frequently attributed to irreversible damage as a side effect of radiation therapy. Stem cell-derived organoid therapy has garnered attention as a promising avenue for resolving this issue. However, Matrigel, a hydrogel commonly used in organoid culture, is considered inappropriate for clinical use due to its undefined composition and immunogenicity. In this study, we aimed to develop a method for culturing collagen-based human salivary gland organoids (hSGOs) suitable for clinical applications and evaluated their therapeutic effectiveness. Human salivary gland stem cells were isolated from the salivary gland tissues and cultured in both Matrigel and collagen. We compared the gene and protein expression patterns of salivary gland-specific markers and measured amylase activity in the two types of hSGOs. To evaluate the therapeutic effects, we performed xenogeneic and allogeneic transplantation using human and mouse salivary gland organoids (hSGOs and mSGOs), respectively, in a mouse model of radiation-induced xerostomia. hSGOs cultured in Matrigel exhibited self-renewal capacity and differentiated into acinar and ductal cell lineages. In collagen, they maintained a comparable self-renewal ability and more closely replicated the characteristics of salivary gland tissue following differentiation. Upon xenotransplantation of collagen-based hSGOs, we observed engraftment, which was verified by detecting human-specific nucleoli and E-cadherin expression. The expression of mucins, especially MUC5B, within the transplanted hSGOs suggested a potential improvement in the salivary composition. Moreover, the allograft procedure using mSGOs led to increased salivation, validating the efficacy of our

approach. This study showed that collagen-based hSGOs can be used appropriately in clinical settings and demonstrated the effectiveness of an allograft procedure. Our research has laid the groundwork for the future application of collagen-based hSGOs in allogeneic clinical trials.

Salivary Gland Volume Predicts Malnutrition in Irradiated Head and Neck Cancer Patients: A Prospective Analysis.

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Xerostomia following (chemo-) radiotherapy (CRT) is common in head and neck (HNC) patients. This prospective study focused on investigating the link between salivary gland function and the onset of malnutrition in HNC patients. Between 2018 and 2020, 61 HNC patients scheduled for curative CRT were prospectively recruited. Nutritional status was assessed by bioelectrical impedance analysis (BIA) and xerostomia was evaluated based on the Common Terminology Criteria for Adverse Events, (CTCAE). Patient-reported outcomes for xerostomia-related symptoms, such as "dry mouth" and "sticky saliva", were also collected. Data were assessed at the beginning of therapy, during treatment, at the end of treatment, and during follow-up. Organs at risk were contoured including the submandibular and parotid glands. Dose-volume parameters were extracted for the mean Dose (Dmean), V15 Gy, V30Gy and V45Gy. No correlation was found between the dosimetric parameters [Dmean, V15Gy (%), V30Gy (%) and V45Gy (%)] and the occurrence of malnutrition [defined by a fat-free mass index (FFMI) <15 kg/m² (♀) and <17 kg/m² (♂) kg/m² and/or body-mass index (BMI) <18.5 kg/m²] at any of the three time points tested. However, the volume of the parotid glands prior to therapy appeared to be related to the development of malnutrition. This effect was not observed with the submandibular glands. A cumulative parotid gland volume of 55.3 cm³ was identified as the threshold for malnutrition at the second follow-up examination. Although none of the dosimetric factors were associated with the development of malnutrition, the baseline parotid gland volume emerged as an independent predictor of malnutrition in head and neck cancer patients with xerostomia.

Trends in Salivary Gland Tissue Engineering: From Stem Cells to Secretome and Organoid Bioprinting.

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Xerostomia or dry mouth are commonly diagnosed in head and neck cancer patients due to salivary gland (SG) epithelial injury after radiotherapy. Regenerative medicine has fetched the opportunity to replace or regenerate the SG epithelia and restore its secretory function. Early adult stem cell transplantation strategies in rodents have recently shown to improve clinical outcomes in radiotherapy-induced xerostomia in Phase 1/2 human trials. Mesenchymal stem cells from adipose tissue are the most promising, although the ones from the labial mucosa, bone marrow, or dental pulp have an attractive therapeutic value after successful findings in ex vivo and in vivo mouse models of SG injury. Emerging approaches using cell-free therapy with cell "extracts", "soups" or secretome components also exhibit favorable outcomes in the same rodent models. When compared to cell-based approaches, extracellular vesicles (EV) from the secretome (i.e., exosomes) can be easily extracted, quantified, and are more stable for long-term storage and use in SG tissue engineering. Additive manufacturing and three-dimensional bioprinting or bioassembly have an important role on generating spheroids or

organoids for cell transplantation to ameliorate SG injury. Moreover, organoids can secrete EV, which may have a therapeutic potential worth to explore in future studies. In this review, we will describe the technological advancements and challenges of these different cell-based and cell-free strategies in SG tissue engineering and regeneration. Impact statement Salivary gland (SG)-like innervated epithelial organoids and the secretome produced from stem cells may constitute feasible therapeutic alternatives to regenerate the SG due to their user-friendly, short-lived, consistent, and scalable additive manufacturing processes. Bioprinting such human SG organoids toward in vitro drug discovery may further reduce the incorporation of animal-derived components to the tissue constructs and minimize the use of animal experimentation in SG regeneration. Despite such advancements, transplantation with human adipose-derived mesenchymal stem cells is the only tissue engineering strategy that has reached Phase 1/2 clinical trials and shown to enlarge the serous SG epithelium and improve salivary flow.

Mesenchymal Stromal/Stem Cell Therapy Improves Salivary Flow Rate in Radiation-Induced Salivary Gland Hypofunction in Preclinical in vivo Models: A Systematic Review and Meta-Analysis.

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Mesenchymal stromal/stem cells (MSCs) have been suggested for salivary gland (SG) restoration following radio-induced salivary gland damage. This study aimed to determine the safety and effectiveness of MSC therapy on radio-induced SG damage and hypofunction in preclinical in vivo studies. PubMed and EMBASE were systematically searched for preclinical in vivo interventional studies evaluating efficacy and safety of MSC treatment following radio-induced salivary gland damage published before 10th of January 2022. The primary endpoint was salivary flow rate (SFR) evaluated in a meta-analysis. The study protocol was published and registered on PROSPERO (www.crd.ac.uk/prosperto), registration number CRD42021227336. A total of 16 preclinical in vivo studies were included for qualitative analysis (858 experimental animals) and 13 in the meta-analysis (404 experimental animals). MSCs originated from bone marrow (four studies), adipose tissue (10 studies) and salivary gland tissue (two studies) and were administered intravenously (three studies), intra-glandularly (11 studies) or subcutaneously (one study). No serious adverse events were reported. The overall effect on SFR was significantly increased with a standardized mean difference (SMD) of 6.99 (95% CI: 2.55-11.42). Studies reported improvements in acinar tissue, vascular areas and paracrine factors. In conclusion, this systematic review and meta-analysis showed a significant effect of MSC therapy for restoring SG functioning and regenerating SG tissue following radiotherapy in preclinical in vivo studies without serious adverse events. MSC therapy holds significant therapeutic potential in the treatment of radio-induced xerostomia, but comprehensive, randomized, clinical trials in humans are required to ascertain their efficacy in a clinical setting.

Gingival-derived mesenchymal stem cell therapy regenerated the radiated salivary glands: functional and histological evidence in murine model.

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Radiotherapy in head and neck cancer management causes degeneration of the salivary glands (SG). This study was designed to determine the potential of gingival mesenchymal stem cells (GMSCs) as a

cell-based therapy to regenerate irradiated parotid SG tissues and restore their function using a murine model. Cultured isolated cells from gingival tissues of 4 healthy guinea pigs at passage 3 were characterized as GMSCs using flow cytometry for surface markers and multilineage differentiation capacity. Twenty-one Guinea pigs were equally divided into three groups: Group I/Test, received single local irradiation of 15 Gy to the head and neck field followed by intravenous injection of labeled GMSCs, Group II/Positive control, which received the same irradiation dose followed by injection of phosphate buffer solution (PBS), and Group III/Negative control, received (PBS) injection only. Body weight and salivary flow rate (SFR) were measured at baseline, 11 days, 8-, 13- and 16-weeks post-irradiation. At 16 weeks, parotid glands were harvested for assessment of gland weight and histological and immunohistochemical analysis. The injected GMSCs homed to degenerated glands, with subsequent restoration of the normal gland histological acinar and tubular structure associated with a significant increase in cell proliferation and reduction in apoptotic activity. Subsequently, a significant increase in body weight and SFR, as well as an increase in gland weight at 16 weeks in comparison with the irradiated non-treated group were observed. The study provided a new potential therapeutic strategy for the treatment of xerostomia by re-engineering radiated SG using GMSCs.

Maintenance of adult stem cells from human minor salivary glands via the Wnt signaling pathway.

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Xerostomia is a salivary gland dysfunction that negatively impacts the life quality of patients; however, there is no effective treatment for xerostomia. Bioengineered organs, generated using stem cells obtained from newborn salivary glands and ligated injury models, are a new organ transplantation strategy that could be feasible for xerostomia treatment. Reconstruction of salivary gland organoids by seed cells obtained from human minor salivary glands will offer theoretical fundamentals and technology support for clinical application and organ regeneration research. Herein, we aimed to propose a new method for culturing and enriching adult human minor salivary gland stem cells in vitro in a three-dimensional (3D) environment via Wnt signaling activation. Obtained and characterized human minor salivary gland stem cells (hMSGSCs) with self-organization ability were 3D-cultured to generate organoids. We examined hMSGSCs proliferation and colony formation using MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assays. Telomerase reverse transcriptase staining, flow cytometry, immunofluorescence assay, RNA isolation, RT-PCR, and qPCR were performed to assess hMSGSCs structure and the function of reconstructive organoids in vitro. hMSGSCs showed typical epithelial-like characteristics, such as positive for CD49f and cell KRT expression. hMSGSCs served as adult stem cells in salivary glands and could differentiate into acinar and duct cells. Upon the addition of Noggin, CHIR99021, and Wnt3A to the 3D culture system, hMSGSCs showed higher LGR5 expression and decreased AMY1B and MUC5B expression. Therefore, the Wnt and bone morphogenetic protein (BMP) pathways are important in regulating hMSGSCs self-organization and differentiation. We showed that the stem cell properties of hMSGSCs in a 3D culture system can be maintained by activating the Wnt signaling pathway and inhibiting the BMP signaling pathway. Our findings contribute new insights on salivary gland organoid generation in vitro.

Association between CUBN gene variants, type 2 diabetes and vitamin D concentrations

in an elderly Greek population.

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Accumulating evidence suggests a potential implication of vitamin D biological network in the pathogenesis of diabetes mellitus. The megalin-cubilin endocytotic system constitutes a key transport structure, with a modulating role in vitamin D metabolism. We aimed to assess the contribution of variants in the CUBN gene to the genetic risk of Type 2 Diabetes Mellitus (T2DM). 95 polymorphisms within CUBN were genotyped in 716 patients with T2DM and 542 controls of Greek origin. Samples were analyzed on Illumina Human PsychArray. Permutation test analysis was implemented to determine statistical significance. Twenty-five-hydroxy-vitamin-D [25(OH)D] levels were measured in a sub-group of participants (n = 276). Permutation analysis associated rs11254375_G/T (pemp = 0.00049, OR = 1.482), rs6602175_G/T (pemp = 0.016, OR = 0.822), rs1801224_G/T (pemp = 0.025, OR = 0.830), rs4366393_A/G (pemp = 0.028, OR = 0.829) and rs7071576_A/G (pemp = 0.04, OR = 1.219) with T2DM. Mean 25(OH)D concentrations were significantly lower in patients with T2DM compared to controls (16.70 ± 6.69 ng/ml vs 18.51 ± 6.71 ng/ml, $p < 0.001$), although both groups were vitamin D deficient. In a further quantitative analysis, rs41301097 was strongly associated with higher 25(OH)D concentrations ($p = 5.233e-6$, $\beta = 15.95$). Our results indicate a potential role of CUBN gene in T2DM genetic susceptibility in the Greek population. These findings may also denote an indirect effect of vitamin D metabolism dysregulation on the pathogenesis of T2DM. Further studies are required to replicate our findings and clarify the complex underlying mechanisms.